CSE221: Algorithms
Lab-01
Fall 2026

Action required for Students

[Google Form] CSE221 Lab Policy:
- The lab policy file can be found here - Link
- Please complete the lab policy acknowledgement form - Link

[Google Form] Codeforces Account Information: (Action required)
- Open an account at https://codeforces.com/
- Please submit your codeforces details here - Link

Students who do not complete this will not be allowed to participate in lab quizzes, and their assignment submissions will not be graded.

Announcements for Students

[Bookmark the Link] Student Drive Folder: Link
The folder contains:
- Lab Files [Fall 2026]
- Lab Policy
- Sample Quiz Checkers
- Environment Setup for Quiz Checker
- Previous Semesters' Lab Quiz Questions

Graded Quiz 01
Date: October 17-18, 2026 (as per your lab class schedule)
Syllabus: Lab 01 content
Take Preparation for both assignment problems and outside assignment problems *[Student Drive Folder Link]*

Assignment-01 Links:

Codeforces Invitation Link:
https://codeforces.com/contestInvitation/8bf630eb2ce286ebc86d1dfbf93a8d8fbec39e83

Alternative Link: https://codeforces.com/contests/722945
[The alternative link will **only work** after you have successfully registered using the first link.]

Self-study: Explore Fast I/O techniques: We must use FAST I/O in our code.

- **Python**: Using `sys.stdin` for faster input handling.
- **C++**: Using `ios_base::sync_with_stdio(false);` and `cin.tie(nullptr)`
- **Java**: Using `BufferedReader` for efficient input reading.
- The fast i/o can reduce program execution time on Online Judge platforms.
- **Fast I/O Example: Python, C++, Java [click Expand to view the code snippet]**

Deadline: `Oct 18, 2026 2:00 PM`
After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading. Also, your assignment submissions will not be evaluated if they are flagged as AI-generated.**

Learning Outcome of this Lab:

- Understand the basics of Online Judge platforms.
- Become familiar with standard Input/Output (I/O) formats used in Online judge
- Learn about common judge verdicts, including Accepted (AC), Wrong Answer (WA), Time Limit Exceeded (TLE), and Runtime Error (RE).
    - For a **1-second** time limit, most OJs allow approximately **10^8 operations**.
    - For python select PyPy over Python if available, as PyPy is faster. However, PyPy consumes more memory than Python.

- Review $O(n^2)$ sorting algorithms which have already been covered in the previous courses such as Bubble Sort, Selection Sort, and Insertion Sort.
    - Learn to sort with multiple parameter
    - Understand the use case of stable sort